E.70 standalone proof review

Statement E.70 (Proved). The full Gaussian pro-three source and its compatible norm lifts. *Let P be the free pro-three group on X, Y, and form the actual relative Gaussian triangle*

$$\begin{aligned} \mathfrak{G} = P \rtimes \langle d \rangle, \qquad d^4 = 1, \quad dXd^{-1} = Y^{-1}, \quad dYd^{-1} = X, \\ x = Xd, \qquad y = Y^{-1}d^2, \qquad z = d. \end{aligned} \tag{E.70.1}$$

It is the quotient of the original quartic triangle obtained by killing its central fourth power and taking the maximal pro-three quotient of its free kernel. Let $\psi : \widehat{F} \to \mathfrak{G}^6$ evaluate the same free word in

$$(x, y), (y, x), (z, x), (x, z), (y, z), (z, y). \tag{E.70.2}$$

Write $C = P/\gamma_3(P)$ in its integral class-two coordinates $(v, c) \in \mathbf{Z}_3^2 \times \mathbf{Z}_3$, with the law (E.69.1) over $\mathbf{Z}_3$. Define

$$A(v, c) = (-v, c - (v_1 + v_2)/2), \qquad B(v, c) = (-Sv, c - (v_1 + v_2)/2). \tag{E.70.3}$$

Then the full six-coordinate image on the actual source $N' = N'_B$ is exactly

$$\begin{aligned} \mathscr{H} &:= \psi(N') = \psi(\widehat{F}') \\ &= \{(g, h) \in P^3 \times P^3 : \overline{h} = (A(\overline{g}_3), B(\overline{g}_2), B(\overline{g}_1))\}. \end{aligned} \tag{E.70.4}$$

Here the bars are the specified maps to C. Projection onto the first three factors is onto P^3, with kernel $\gamma_3(P)^3$ in the last three factors. Moreover

$$\mathscr{H}^{\mathrm{ab}} \simeq \mathbf{Z}_3^{12}. \tag{E.70.5}$$

The six weight-three coordinates in this abelianization are retained in the following common-source assertion.

Keep exactly the finite E.66 map $\Psi : N' \to Z$ and the unchanged E.67 Mordell map $\Theta_M : N' \to \mathcal{M}$. The full common image is

$$(\Psi, \Theta_M, \psi)(N') = Z \times \mathcal{M} \times \mathscr{H}. \tag{E.70.6}$$

Every one of the 3456 finite combinations in E.69 admits a compatible lift through the entire Gaussian pro-three group and all its characteristic finite quotients. For each of the four Gaussian actions, the lift can be chosen to be the actual automorphism $\mathrm{Inn}(x^j)$, $0 \le j < 4$, with both frames in P and the original six word equations retained. Thus the finite E.69 frame and all the specified E.66 and Mordell values can be retained by one profinite word.

For one chosen full target over an actual E.17 origin f_i, and one such correction $u \in N'$, its entire word fibre inside $f_i N'$ is

$$f_i u\big(N' \cap \ker \Psi \cap \ker \Theta_M \cap \ker \psi\big). \tag{E.70.7}$$

The Mordell factor here remains its specified order-54 quotient. The statement supplies the full Gaussian tower comparison, with the displayed finite conditions; it assigns no full-GT lift or additional genus-one compatibility equation.

Proof. We first prove the exact six-coordinate image, then identify its complete mod-three abelianization and its actual common quotient with the retained source.

The actual free-kernel maps. The presentation (E.70.1) is the exact E.64 quartic presentation, followed by its indicated quotient. The source generators still have product $xyz = 1$, and direct free reduction gives

$$y^2 = z^4 = 1, \qquad x^4 = c := [X, Y^{-1}], \qquad x^2 y = X, \qquad z^2 y = Y. \tag{E.70.8}$$

In particular x, y generate the whole target $\mathfrak{G}$. All six maps (E.70.2) are actual continuous free-source maps. On $\widehat{F}'$ they have deck exponent zero and therefore take values in P^6.

The maps A, B preserve the law of C because their linear parts have determinant one and their central corrections are linear. For the three ordered pairs of evaluations

$$(3, 4, A), \qquad (2, 5, B), \qquad (1, 6, B), \tag{E.70.9}$$

there are exact intertwining identities for conjugation by each of the two free-source generators. They can be checked without any finite reduction: conjugation by (w, d^e) on C sends

$$(v, c_0) \longmapsto (S^e v, c_0 + \det(\log(w)_{\mathrm{ab}}, S^e v)). \tag{E.70.10}$$

Substitution in (E.70.3) gives all six polynomial identities over $\mathbf{Z}[1/2]$. On the normal generator $[x_{\mathrm{s}}, y_{\mathrm{s}}]$ of $\widehat{F}'$, the three comparisons, in the same order as (E.70.9), are

	first class-two value	second class-two value
A	$(-1, -1, -1/2)$	$(1, 1, 1/2)$
B	$(-1, -1, -1/2)$	$(1, -1, 1/2)$
B	$(1, 1, 1/2)$	$(-1, 1, -1/2)$.

(E.70.11)

The six covariance identities propagate these three identities under all source conjugations, products, inverses and limits. Since the closed normal closure of that commutator is $\widehat{F}'$, they prove the upper inclusion in (E.70.4) on the whole required domain. The exact polynomial and reduced-word checks are saved in `bc_gaussian_pro3_maps.py`.

The E.67 Gaussian image on N' gives all six independent coordinates of the first three copies of $P/\Phi(P)$. Hence its closed image in P^3 is onto. Indeed, a proper closed subgroup would have a proper image in some finite three-group, and a maximal subgroup of that finite group would give a missing mod-three abelian character. The same argument proves surjectivity onto the last three copies of P, and onto the pair of coordinates $(1, 4)$, whose E.67 values are independently u and $-t$.

The three full kernel factors, with their source membership. Here $x_{\mathrm{s}}, y_{\mathrm{s}}$ are the original free-source generators and $z_{\mathrm{s}} = (x_{\mathrm{s}} y_{\mathrm{s}})^{-1}$. Literal free reduction in (E.70.1) gives the complete six-tuples

$$\begin{aligned} \psi(x_{\mathrm{s}}^4) &= (c, 1, 1, c, 1, 1), \\ \psi(z_{\mathrm{s}}^4) &= (1, 1, 1, 1, c, dcd^{-1}). \end{aligned} \tag{E.70.12}$$

The elements $x_{\mathrm{s}}^4, z_{\mathrm{s}}^4$ are used in $\widehat{F}$; no assertion that they belong to N' is needed. The subgroup N is normal in the actual modular source, so N' is normal in $\widehat{F}$. Consequently their commutators with any $h \in N'$ do belong to N'.

For any $a \in P$, use the last-three-factor surjection to choose $h \in N'$ with those coordinates $(1, a, 1)$. Then $[z_{\mathrm{s}}^4, h]$ has only its fifth coordinate nontrivial, equal to $[c, a]$. Choosing $(1, 1, a)$ gives only $[dcd^{-1}, a]$ in the sixth coordinate. Finally choose $h \in N'$ with first coordinate one and fourth coordinate a, using the proved $(1, 4)$ surjection. The element $[x_{\mathrm{s}}^4, h]$ has only its fourth coordinate nontrivial, equal to $[c, a]$.

The closed normal subgroup generated by $[c, X], [c, Y]$ is exactly $\gamma_3(P)$. Both words belong to $\gamma_3(P)$. In their quotient c is central, because X, Y generate P. The relation $c = [X, Y^{-1}]$ then makes $[X, Y]$ central as well; the quotient is class two, giving the reverse inclusion. Conjugating c by d gives the same conclusion since $\gamma_3(P)$ is characteristic. The kernel of projection of $\psi(N')$ onto the first three factors is normal under the full last-three-factor image P^3. The three isolated families above therefore supply every element of $\gamma_3(P)^3$ in that kernel. The upper inclusion already bounds the kernel by this group. Together with the first-three-factor surjection this proves equality in (E.70.4). It also proves equality with $\psi(\widehat{F}')$.

All twelve abelian characters and a finite detector. Put $c_0 = [X, Y]$. The following continuous endomorphisms are actual automorphisms of P:

	X	Y
$\widetilde{A}$	$X^{-1}c_0^{-1/2}$	$Y^{-1}c_0^{-1/2}$
$\widetilde{B}$	$Yc_0^{-1/2}$	$X^{-1}c_0^{-1/2}$.

(E.70.13)

The powers are continuous pro-three powers. Their Frattini matrices are $-I$ and $-S$, so both maps are onto. They are isomorphisms: the iterated Frattini subgroups give cofinal characteristic finite quotients, on which these surjections are bijective, and their kernels consequently have trivial intersection. Their maps on C are exactly A, B.

For the pairs $(1, 6), (2, 5), (3, 4)$, apply the appropriate lift in (E.70.13) to the first entry. The second entry times its inverse lies in $\gamma_3(P)$. This gives actual split descriptions of the three fibre products as $\gamma_3(P) \rtimes P$, with conjugation through the indicated lift. Hence

$$\mathscr{H}^{\mathrm{ab}} \simeq \left(P^{\mathrm{ab}} \oplus \gamma_3(P)/\gamma_4(P)\right)^3 \simeq (\mathbf{Z}_3^2 \oplus \mathbf{Z}_3^2)^3. \tag{E.70.14}$$

For completeness the last rank statement follows from the actual class-three collection coordinates

$$X^a Y^b c_0^e [c_0, X]^u [c_0, Y]^v, \qquad a, b, e, u, v \in \mathbf{Z}_3. \tag{E.70.15}$$

Commutator collection gives existence, first for integer words and then by compactness. Uniqueness is detected by the corresponding upper-unitriangular 4-by-4 matrices: the two linear coefficients detect a, b, the weight-two coefficient detects e, and the two independent weight-three coefficients detect u, v. Thus these are ordinary integral lower-central coordinates, including at the prime three.

Use the characteristic finite quotient

$$P_3 = P/\overline{\gamma_4(P)\gamma_2(P)^3 P^9}. \tag{E.70.16}$$

In (E.70.15) its coordinate ranges are $(9, 9, 3, 3, 3)$, so $|P_3| = 3^7$. The common class-two quotient has order 3^5. The image S_3 of $\mathscr{H}$ in P_3^6 is the corresponding threefold fibre product, of order 3^{27}. The splits above reduce to

$$S_3 \simeq (P_3 \times \mathbf{F}_3^2)^3, \qquad \mathscr{H}/\Phi(\mathscr{H}) \xrightarrow{\sim} S_3/\Phi(S_3) \simeq \mathbf{F}_3^{12}. \tag{E.70.17}$$

One can see the last map directly: in each pair retain the first entry's two abelian coordinates modulo three and the two weight-three coordinates of the second entry times the inverse of the specified lift of the first. Formula (E.70.15) shows that these are precisely the four reductions of its integral abelianization coordinates.

The finite certificate implements all these maps. Write a strictly upper-unitriangular matrix as $(a_{12}, a_{23}, a_{34}, a_{13}, a_{24}, a_{14})$. The four pairs of generator images in $U_4(\mathbf{F}_3)$ are

	X	Y
1	$(1,1,1,0,0,0)$	0
2	0	$(1,1,1,0,0,0)$
3	$(1,1,0,0,0,0)$	$(0,0,1,0,0,0)$
4	$(1,0,0,0,0,0)$	$(0,1,1,0,0,0)$.

(E.70.18)

Their 2187 collected values are all distinct. Four central characters, evaluated on $X^3, Y^3, [c_0, X], [c_0, Y]$, have rank four. Each matrix acts faithfully on its three affine coordinates, and the product action has degree 108. Induction through the actual order-four deck automorphism gives degree 432, and all six literal evaluations give degree 2592. The program checks the deck automorphism on every element and both generating products, with the right-action convention $p(gh) = p(h)[p(g)]$. A separate character certificate decodes the actual six matrices and checks all twelve values of (E.70.17), including every saved word DAG; their rank is twelve.

The exact common image with the retained finite source. Adjoin those six actions and the six unchanged Mordell actions to the actual 5223-point E.66 source. The source has degree 7869. All 97 literal N words are compared with the E.67 export. The complete Schreier calculation first computes the image of N, and then its derived subgroup, which is the image of N'. Only after this step does it retain the five coordinate-zero distribution blocks. The resulting degree is 7495. The independent full Schreier replays give

actual retained projection	order
S_3	3^{27}
$\mathcal{M} \times S_3$	3^{30}
Z and S_3	$\lvert Z \rvert\, 3^{27}$
Z and $\mathcal{M}$ and S_3	$\lvert Z \rvert\, 3^{30}$.

(E.70.19)

Each equality is for the actual common-domain map; it follows from the complete source certificates, not a comparison of independent projection orders.

Since (E.70.17) is the full Frattini quotient, the closed image of N' in $\mathcal{M} \times \mathscr{H}$ is the whole pro-three product. A common quotient of that product with finite Z is a finite three-group. Any nontrivial such quotient has a nontrivial cyclic quotient of order three. Its character factors through the specific map

$$\mathcal{M} \times \mathscr{H} \longrightarrow \mathcal{M} \times S_3 \longrightarrow (\mathcal{M} \times S_3)/\Phi(\mathcal{M} \times S_3). \tag{E.70.20}$$

This would give a common cyclic quotient in the last actual product in (E.70.19), which is impossible. Goursat's exact quotient construction therefore proves (E.70.6).

Compatible full Gaussian frames for every retained finite value. For $j = 0, 1, 2, 3$, let $\alpha_j = \mathrm{Inn}(x^j)$. It fixes x and its primitive fourth power. The elements y and $\alpha_j(y)$ have order two and the same image in the deck group. Coprime complement conjugacy gives a frame $h_y \in P$. The element $h_z = x^j d^{-j}$ is already a frame for $\alpha_j(z)$. Their possible class-two central coordinates may be prescribed independently. Here is the exact lifting argument: for an element fixed by y or d in C, lift it to P. Its conjugation sends the order-two or order-four complement to another complement differing within $\gamma_3(P)$. Conjugacy within that pro-three kernel corrects the lift. This proves the surjections of the corresponding fixed centralizers. For a finite three-group kernel, the conjugacy assertion follows by induction through a central

invariant elementary-abelian subgroup and averaging the resulting cocycle, since two and four are units. For the pro-three kernel, the finite conjugator sets are nested nonempty compact sets. This proves the claimed conjugacy and centralizer maps in the full groups.

The fixed centralizer in C consists of its central coordinate in each case: $-I-I$ and $S-I$ are invertible over $\mathbf{Z}_3$. We can therefore choose the frames to have precisely

j	$\overline{h}_y$	$\overline{h}_z$
0	$(0,0,q)$	$(0,0,0)$
1	$(1/2,1/2,q)$	$(1,0,1/4)$
2	$(1,0,q)$	$(1,-1,0)$
3	$(1/2,-1/2,q)$	$(0,-1,-1/4)$,

$q \in \mathbf{Z}_3$. (E.70.21)

The abelian coordinates follow by conjugating the literal elements (E.70.1). Substitution of these coordinates in the six values (E.69.12) satisfies all three integral graph identities (E.70.4). The four polynomial checks hold for the unrestricted parameter q and are included in the literal-map certificate. Thus the entire six-tuple lies in $\mathscr{H}$, while its full group action is the chosen actual α_j. Reducing (E.70.21) modulo three gives exactly the twelve E.69 framed targets.

For an actual origin f_i, its six Gaussian values lie in $\mathscr{H}$ by (E.70.4). The correction from them to any of the chosen full norm tuples is therefore an element of $\mathscr{H}$. Choose its simultaneous N' preimage in (E.70.6), retaining the prescribed E.66 correction and the unique finite Mordell frame. This proves the compatible whole-tower lift of every E.69 finite combination. The exact kernel of that common homomorphism gives the full word fibre (E.70.7). □